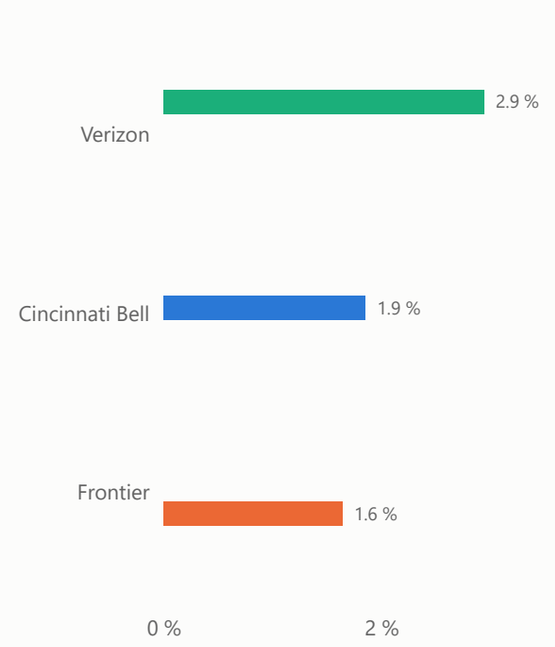
5 of 237 analyzed FTTH units (2.1%) are at risk

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Units in Scope	Analyzed Units	At Risk	Severe
258	237	5	1

% of analyzed units at risk, by ISP



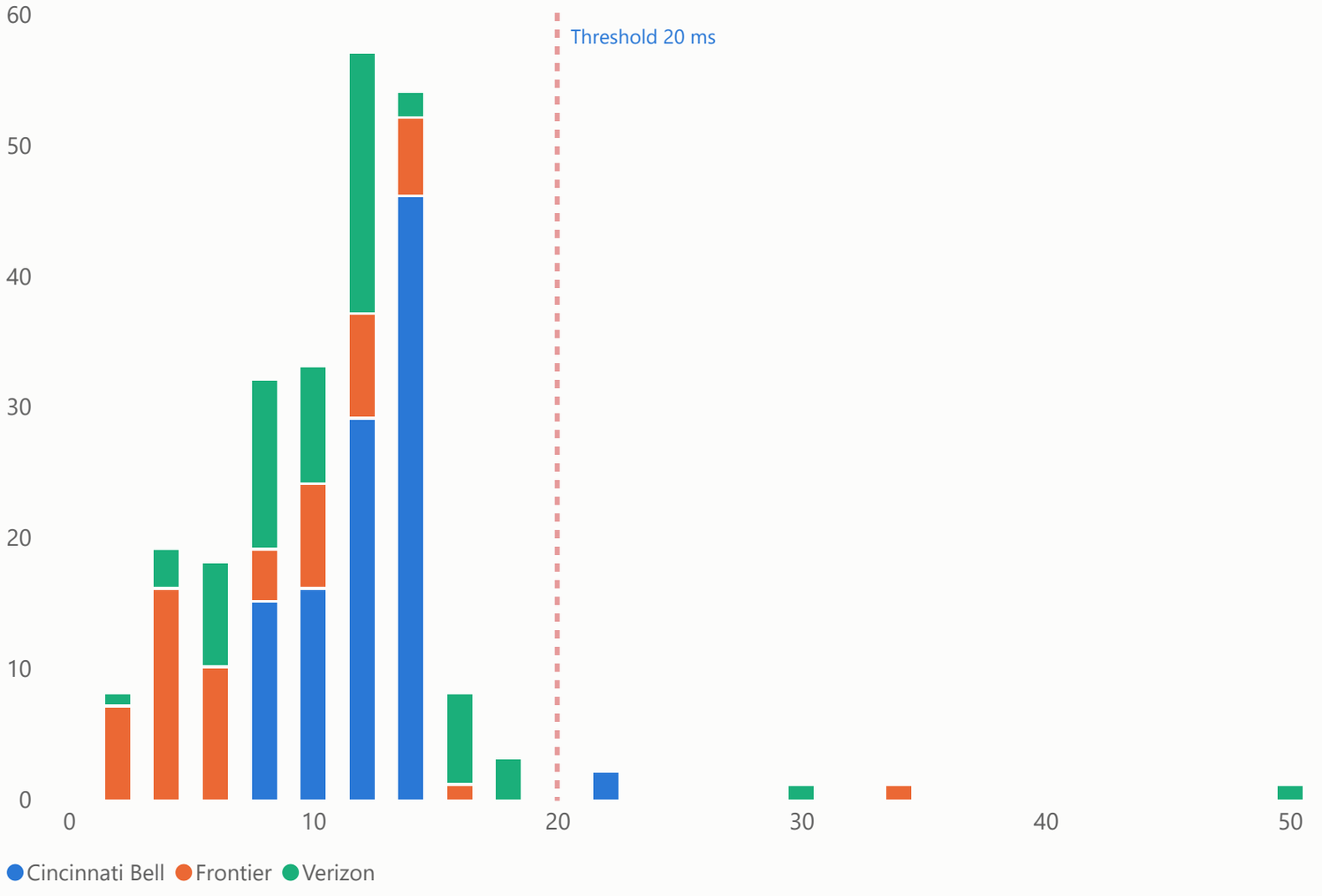
Analyzed and at-risk units, by ISP and state

isp	Analyzed Units	At Risk Units	% At Risk
Cincinnati Bell	**108**	**2**	**1.9 %**
OH	95	2	2.1 %
KY	13	0	0.0 %
Verizon	**68**	**2**	**2.9 %**
MD	11	1	9.1 %
NY	11	1	9.1 %
DE	1	0	0.0 %
MA	9	0	0.0 %
NJ	6	0	0.0 %
PA	12	0	0.0 %
RI	2	0	0.0 %
VA	16	0	0.0 %
Frontier	**61**	**1**	**1.6 %**
CA	18	1	5.6 %
AL	1	0	0.0 %
CT	2	0	0.0 %
FL	20	0	0.0 %
IN	2	0	0.0 %
TX	18	0	0.0 %
Total	**237**	**5**	**2.1 %**

5 of 237 analyzed units exceed 20 ms P95 latency

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P95 latency distribution of analyzed units vs the 20 ms threshold



Analyzed Units

237

Latency Flags

5

Units above the 20 ms threshold

unit_id	isp	state	p95_latency_ms
4190285	Verizon	MD	50.0
24840041	Frontier	CA	35.8
25314177	Verizon	NY	30.5
24733221	Cincinnati Bell	OH	24.0
943226	Cincinnati Bell	OH	23.1

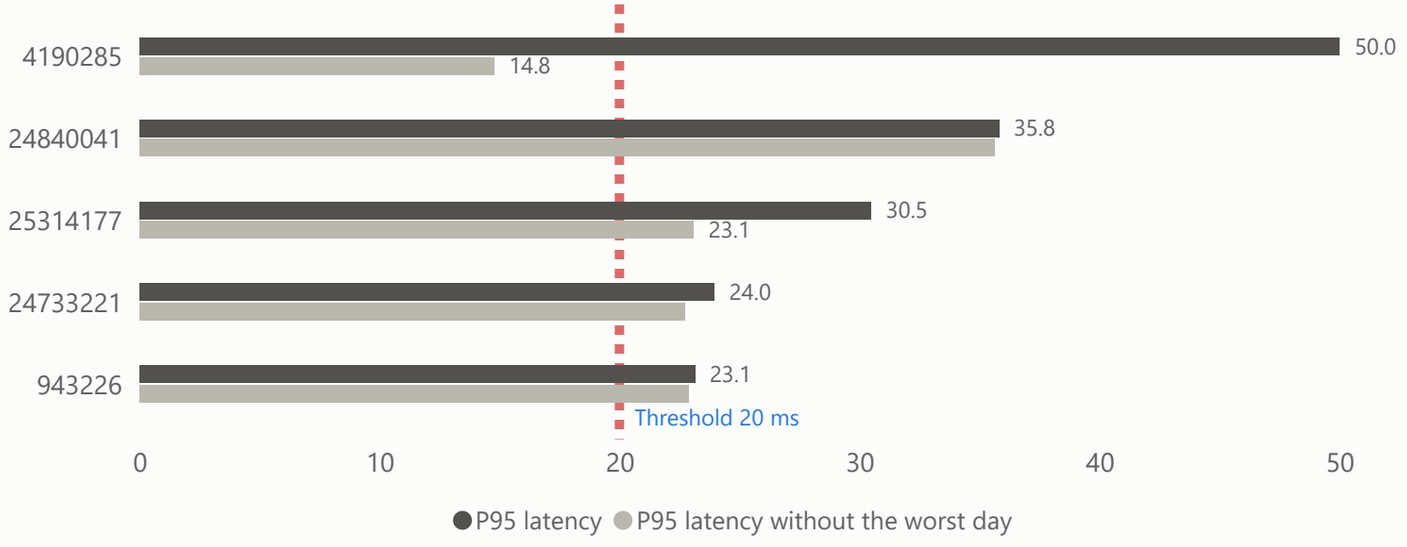
5 units are at risk: 4 persistent, 1 episodic

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At-risk units and their latency pattern

unit_id	isp	state	p95_latency_ms	p95_latency_without_worst_day_ms	n_days_above_threshold	Latency Pattern
4190285	Verizon	MD	50.0	14.8	1	Episodic
24840041	Frontier	CA	35.8	35.6	16	Persistent
25314177	Verizon	NY	30.5	23.1	9	Persistent
24733221	Cincinnati Bell	OH	24.0	22.7	7	Persistent
943226	Cincinnati Bell	OH	23.1	22.9	9	Persistent

P95 latency vs P95 without the worst day, at-risk units



Persistent Flags

4

Episodic Flags

1

How to read this page

Persistent: the P95 stays above 20 ms even after removing the worst day.

Episodic: the P95 falls to 20 ms or below once the worst day is removed.

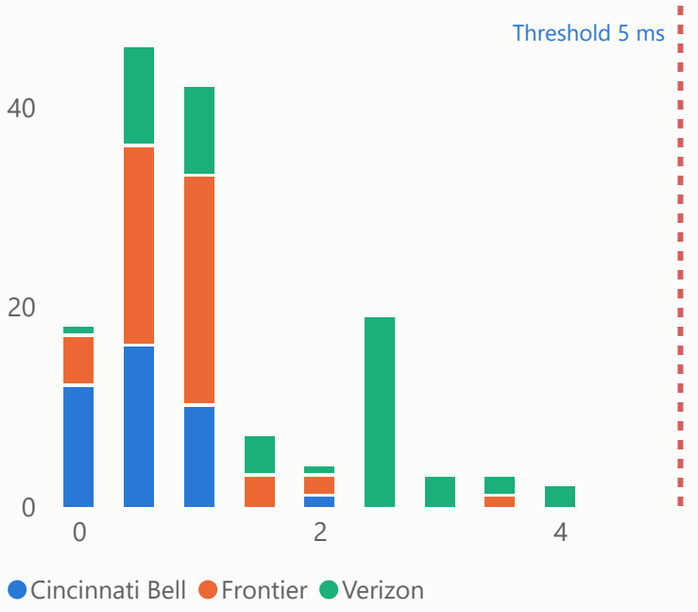
The label is descriptive only (D-10): the risk rule does not change, and all 5 units stay at risk.

Unit 4190285 is also the only severe unit: its P95 of 50.0 ms is an interpolation, and it is the episodic case.

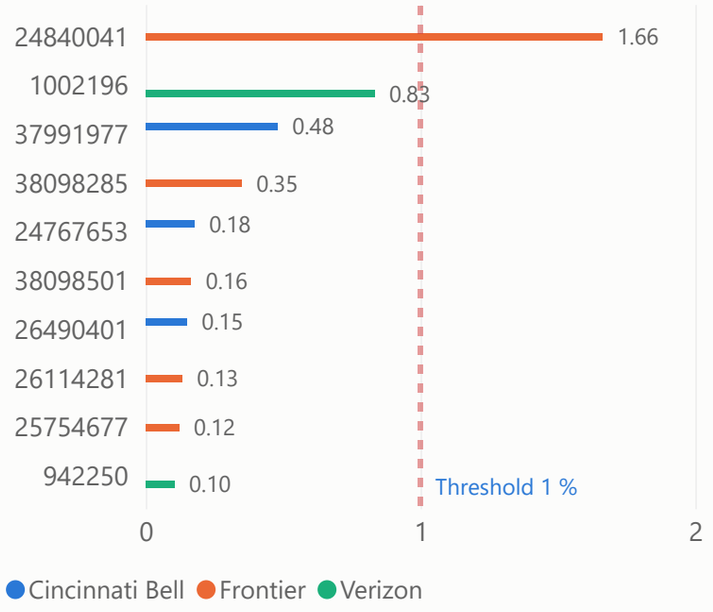
No measurable unit exceeds 5 ms P95 jitter

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P95 jitter of units with enough peak records



Ten highest packet loss values at peak



Units without measurable jitter

93

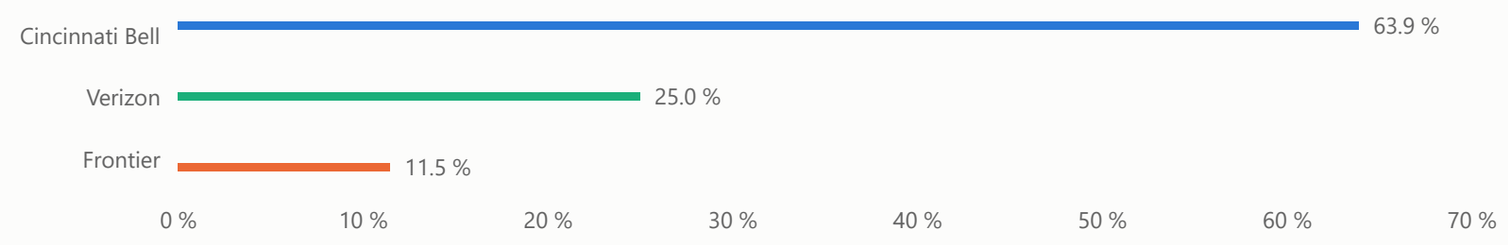
Jitter Flags

0

Loss Flags

1

Share of analyzed units without measurable jitter, by ISP



How to read this page

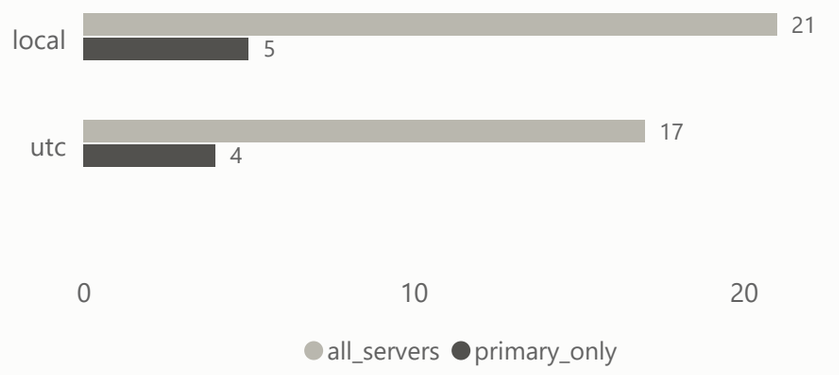
Jitter is left empty (never 0) when a unit has fewer than 50 peak jitter records (D-07). 93 of 237 units cannot be assessed for jitter, 69 of them from Cincinnati Bell.

No measurable unit exceeds 5 ms P95 jitter; the highest is 4.48 ms (unit 4190285).

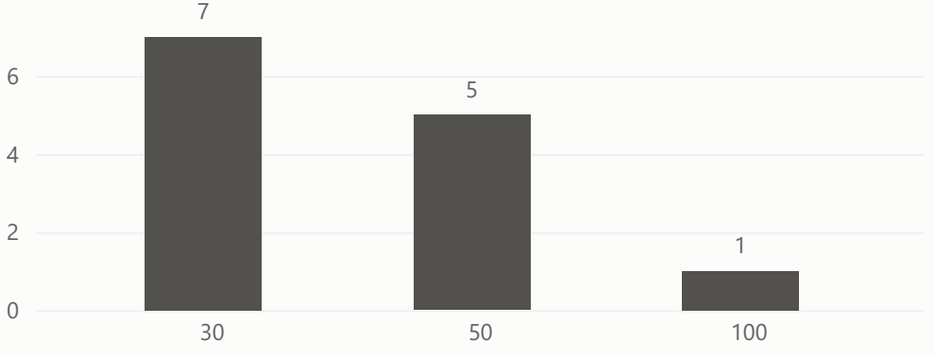
Only unit 24840041 (Frontier, 1.66 %) exceeds 1 % packet loss; the next highest is 0.83 %.

Base case: 5 units at risk; using all test servers raises it to 21

Units at risk by time reading and server rule



Units at risk by minimum peak records (N_MIN)



N_MIN (minimum peak records)

30 | **50** | 100

Jitter threshold (ms)

3 | 4 | **5**

Units at risk by loss threshold (rows, %) and latency threshold (columns, ms)

Loss threshold (%)	20	30	50
0.05	22	20	19
0.10	14	12	11
0.50	6	4	3
1.00	5	3	2

How to read this page

The base case (local time, primary server, N_MIN 50, thresholds 20 ms / 5 ms / 1 %) gives 5 units at risk.

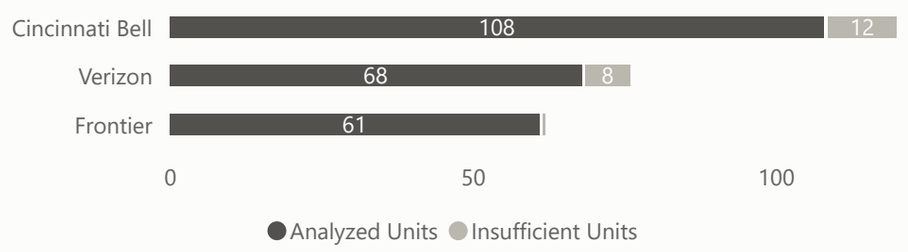
The server rule has the largest effect: 5 with the primary server, 21 with all servers. N_MIN moves the count between 7 and 1. Across the full threshold grid it ranges from 1 to 30.

Select N_MIN and the jitter threshold to explore the grid. The base-case cell is loss 1 / latency 20 with N_MIN 50 and jitter 5.

258 units in scope: 237 analyzed, 21 set aside

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Analyzed and insufficient units by ISP



Data cleaning log (all ISPs): records at each rule

scope	rule	count
udpjitter	duplicate_conflicting_groups	1
udpjitter	duplicate_conflicting_rows_dropped	2
udpjitter	duplicate_exact_extra_rows_removed	4
udpjitter	ftth_rows_read	152,730
udpjitter	invalid_dtime_not_parseable	0
udpjitter	invalid_jitter_null_or_negative	0
udpjitter	rows_after_cleaning	152,724
udpjitter	rows_in_analysis_period	152,730
udplatency	duplicate_conflicting_groups	0
udplatency	duplicate_conflicting_rows_dropped	0
udplatency	duplicate_exact_extra_rows_removed	9
udplatency	ftth_rows_read	172,800
udplatency	invalid_dtime_not_parseable	0
udplatency	invalid_loss_pct_outside_0_100	0
udplatency	invalid_rtt_avg_null_or_negative	0
udplatency	invalid_successes_plus_failures_le_0	0
udplatency	rows_after_cleaning	172,791
udplatency	rows_in_analysis_period	172,800

How the result was built

Scope: 258 FTTH units; analysis period 2022-09-12 to 2022-10-31 (D-01).
Peak window: 19:00-22:59 local time (D-03).
One server per unit: the primary server of the latency file, used for latency, loss and jitter (D-04, D-06).
Metrics per unit: P95 latency, P95 jitter and packet loss (ratio of sums) over peak-window records.
A unit is analyzed with at least 50 peak records (D-02). Jitter needs at least 50 peak jitter records, otherwise it stays empty, never 0 (D-07).
Risk flag: P95 latency > 20 ms, P95 jitter > 5 ms or loss > 1 % (strictly greater). Severe: > 50 ms, > 15 ms or > 10 %.
The Python pipeline and an independent Power Query rebuild agree on all 258 units (0 mismatches).

Limits

Single period (2022) and a non-random sample of volunteer units; results do not generalize to all FTTH subscribers.
220 of 258 units (85 %) are in the US Eastern time zone.
Risk thresholds are project-defined; their sensitivity is shown on page 5.
The local-time reading of dtime rests on indirect evidence; the UTC reading is reported as a sensitivity.
Jitter is evaluated for 144 of 237 analyzed units (61 %).
The count at risk depends on N_MIN, and some flags are episodic (page 3).